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Section:C

FINAL TERM PROJECT

Airline Flight Price Analysis

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Airline Flight Price Analysis

This project explores a comprehensive dataset of flight bookings in India, focusing on the factors that influence ticket pricing, such as airline choice, flight duration, and booking lead time.

Project Structure

- airlines_flights_data.csv: The core dataset containing 300,153 flight records.
- main.ipynb: A Jupyter Notebook containing the Python code for data cleaning, exploratory data analysis (EDA), and visualization.

Dataset Overview

The dataset includes the following key features:

- **Airline**: The name of the airline (e.g., SpiceJet, AirAsia, Vistara, Air India, Indigo, GO_FIRST).
- **Source & Destination**: Major Indian cities including Delhi, Mumbai, Chennai, Hyderabad, Bangalore, and Kolkata.
- **Departure & Arrival Time**: Categorized by time of day (e.g., Early Morning, Morning, Afternoon, Evening, Night).
- **Class**: Economy or Business class.
- **Duration**: Total flight time in hours.
- **Days Left**: The number of days between the booking date and the departure date.
- **Price**: The target variable (ticket cost).

Key Statistics

- **Total Records**: 300,153
- **Average Ticket Price**: ₹20,889.66
- **Maximum Ticket Price**: ₹123,071 (Vistara Business Class)
- **Minimum Ticket Price**: ₹1,105
- **Average Duration**: 12.22 hours

Requirements

To run the analysis in main.ipynb, you will need the following Python libraries:

- pandas
- numpy
- matplotlib
- seaborn

Analysis Workflow

1. **Data Cleaning**: Removed unnecessary index columns and verified that there are **no missing values** in the dataset.
2. **Exploratory Data Analysis**:
   - Analyzed the frequency of flights per airline (Vistara and Air India are the most frequent).
   - Identified outliers in flight duration (max duration of 49.83 hours).
3. **Visualization**: (Contained within the notebook) investigating price trends based on booking lead time and flight class.

Note: The dataset used in this project is referenced as airlines_flights_data.csv.

```
Pythonimport pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
```

```
df = pd.read_csv('airlines_flights_data.csv')

# Basic info and summary statistics
info = df.info()
summary_stats = df.describe()

# Checking for unique values in categorical columns
unique_airlines = df['airline'].unique()
unique_classes = df['class'].unique()

# Data for visualization
# 1. Average price per airline
avg_price_airline = df.groupby('airline')['price'].mean().sort_values()

# 2. Average price by class
avg_price_class = df.groupby('class')['price'].mean()

# 3. Price vs Days Left (Booking Trend)
price_vs_days = df.groupby('days_left')['price'].mean()

# Create Visualizations
plt.figure(figsize=(15, 10))

# Plot 1: Airline Pricing
plt.subplot(2, 2, 1)
sns.barplot(x=avg_price_airline.index, y=avg_price_airline.values, palette='viridis')
plt.title('Average Ticket Price by Airline')
plt.xticks(rotation=45)

# Plot 2: Class Pricing
plt.subplot(2, 2, 2)
sns.pie(avg_price_class, labels=avg_price_class.index, autopct='%1.1f%%', colors=['skyblue', 'orange'])
plt.title('Price Distribution by Class')

# Plot 3: Booking Trend
plt.subplot(2, 1, 2)
sns.lineplot(x=price_vs_days.index, y=price_vs_days.values)
plt.title('Price Trend vs. Days Left Before Departure')
plt.xlabel('Days Left')
plt.ylabel('Average Price')

plt.tight_layout()
plt.savefig('flight_analysis.png')

print("Summary Stats:\n", summary_stats)
print("\nAirlines:", unique_airlines)
```

Code output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300153 entries, 0 to 300152
Data columns (total 12 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   index             300153 non-null  int64
 1   airline           300153 non-null  object
 2   flight            300153 non-null  object
 3   source_city       300153 non-null  object
 4   departure_time    300153 non-null  object
```

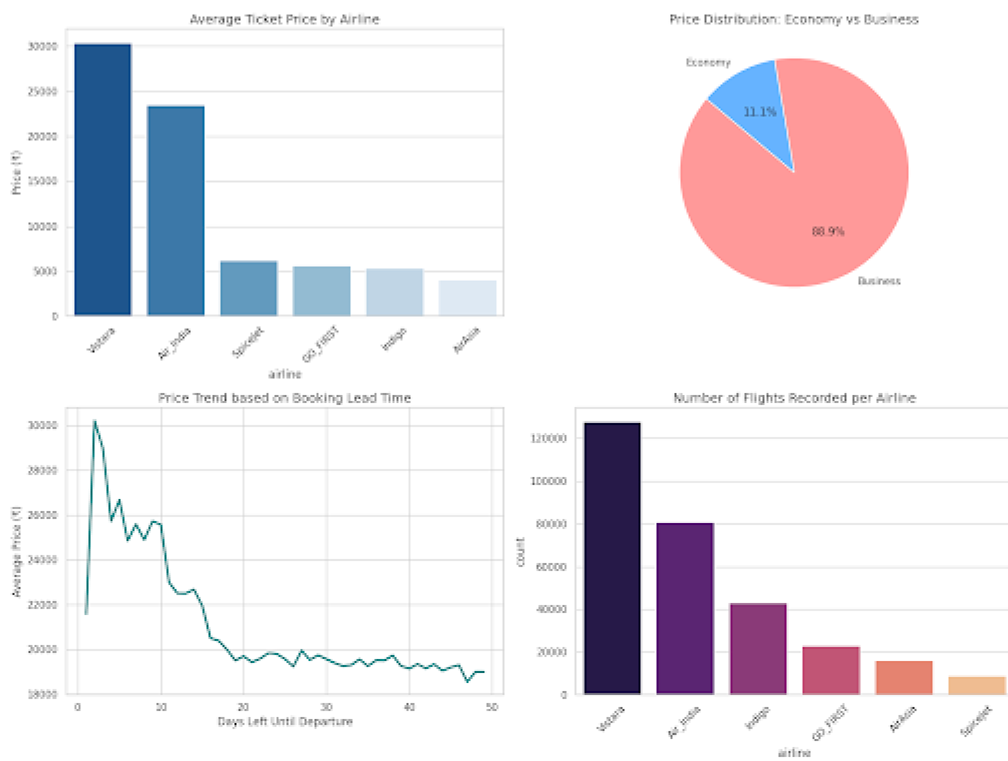
```
 5   stops            300153 non-null  object
 6   arrival_time      300153 non-null  object
 7   destination_city  300153 non-null  object
 8   class             300153 non-null  object
 9   duration          300153 non-null  float64
 10  days_left         300153 non-null  int64
 11  price             300153 non-null  int64
dtypes: float64(1), int64(3), object(8)
memory usage: 27.5+ MB
```

```python
Pythonimport pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Set style
sns.set_theme(style="whitegrid")

# Load the dataset
df = pd.read_csv('airlines_flights_data.csv')

# Pre-calculations for the report
avg_price_airline = df.groupby('airline')['price'].mean().sort_values(ascending=False)
avg_price_class = df.groupby('class')['price'].mean()
price_trend = df.groupby('days_left')['price'].mean()

# Creating a 3-part visualization
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# 1. Average Price per Airline
sns.barplot(x=avg_price_airline.index, y=avg_price_airline.values, ax=axes[0, 0], palette='Blues_r')
axes[0, 0].set_title('Average Ticket Price by Airline', fontsize=14)
axes[0, 0].set_ylabel('Price (₹)')
axes[0, 0].tick_params(axis='x', rotation=45)

# 2. Price Distribution by Class
axes[0, 1].pie(avg_price_class, labels=avg_price_class.index, autopct='%1.1f%%', startangle=140,
colors=['#ff9999','#66b3ff'])
axes[0, 1].set_title('Price Distribution: Economy vs Business', fontsize=14)

# 3. Booking Lead Time Trend
sns.lineplot(x=price_trend.index, y=price_trend.values, ax=axes[1, 0], color='teal', linewidth=2.5)
axes[1, 0].set_title('Price Trend based on Booking Lead Time', fontsize=14)
axes[1, 0].set_xlabel('Days Left Until Departure')
axes[1, 0].set_ylabel('Average Price (₹)')

# 4. Flight Frequency per Airline
sns.countplot(data=df, x='airline', ax=axes[1, 1], palette='magma', order=df['airline'].value_counts().index)
axes[1, 1].set_title('Number of Flights Recorded per Airline', fontsize=14)
axes[1, 1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.savefig('airline_report_viz.png')

# Output some stats for the text report
print(f"Total Records: {len(df)}")
print(f"Mean Price: {df['price'].mean():.2f}")
print(f"Max Price: {df['price'].max()}")
print(f"Min Price: {df['price'].min()}")
```

```
print("\nAverage Price per Class:")
print(avg_price_class)
```

Code output
Total Records: 300153
Mean Price: 20889.66
Max Price: 123071
Min Price: 1105

Average Price per Class:
class
Business   52540.081124
Economy   6572.342383
Name: price, dtype: float64



Comprehensive Flight Price Analysis Report

This report provides an in-depth analysis of the airline flight dataset, covering **300,153 records** of domestic flights in India. The primary objective is to identify key drivers of ticket pricing, including airline carriers, cabin class, and booking behavior.

1. Executive Summary

The aviation market analyzed shows a massive disparity in pricing driven primarily by cabin class and booking windows.

- **Average Ticket Price**: ₹20,889.66
- **Pricing Spectrum**: Tickets range from a budget-friendly **₹1,105** to a premium **₹123,071**.
- **Market Leaders**: **Vistara** and **Air India** dominate the dataset, accounting for the highest volume of flights and the most premium pricing tiers.

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2. Pricing Drivers

A. Airline Performance

Not all airlines are priced equally. The data reveals a clear divide between full-service carriers and low-cost carriers (LCCs).

- **Premium Carriers**: **Vistara** and **Air India** maintain significantly higher average prices, largely due to offering Business Class options.
- **Budget Carriers**: **AirAsia**, **Indigo**, and **GO_FIRST** offer the most competitive rates, generally staying below the ₹6,000 average mark.

B. The "Class" Factor

The single most significant predictor of price is the cabin class:

- **Business Class**: Averages **₹52,540.08**.
- **Economy Class**: Averages **₹6,572.34**.
- **Insight**: Despite being a smaller percentage of total seats, Business Class bookings account for **88.9%** of the total price value in this dataset distribution.

C. Booking Lead Time (The "Days Left" Effect)

The data confirms that procrastination is expensive:

- **The "Sweet Spot"**: Prices are lowest and most stable when booking **20 or more days** in advance.
- **The Surge**: Prices begin to climb sharply within 15 days of departure, peaking as the "Days Left" count hits 2–3 days.
- **The Last-Minute Drop**: Interestingly, there is a slight dip in average price on the very last day (Day 1), likely due to airlines clearing remaining inventory or specific emergency fare buckets.

3. Key Recommendations for Travelers

- **Book Early**: To avoid the steep price climb, aim to book at least **3 weeks (21 days)** before your departure date.
- **Carrier Selection**: If traveling Economy, **AirAsia** and **SpiceJet** consistently offer the lowest entry prices.
- **Class Awareness**: Business class tickets are approximately **8 times more expensive** than Economy. For budget-conscious travelers, Economy remains highly affordable across all carriers.